We create chemistry

Safety data sheet

1. Substance/preparation and manufacturer/supplier identification

Product name:
Valendo

Use: crop protection product, insecticide

Manufacturer/supplier:
BASF (Thai) Limited
23rd Floor, Emporium Tower, 622, Sukhumvit 24 Rd.,
Klongton, Klongtoey, Bangkok 10110, THAILAND
Telephone: +66 2624-1999
Telefax number: +66 2664-9254
E-mail address: Thailand-SDS-info@basf.com

Emergency information:
International emergency number:
Telephone: +49 180 2273-112

2. Hazard identification

Classification according to UN GHS 2009

Classification of the substance and mixture:
Acute toxicity: Cat.3 (oral)
Acute toxicity: Cat.2 (Inhalation - mist)
Skin corrosion/irritation: Cat.3
Specific target organ toxicity — single exposure (Central nervous system): Cat.2
Specific target organ toxicity — repeated exposure (Central nervous system): Cat.2
Hazardous to the aquatic environment - acute: Cat.1
Hazardous to the aquatic environment - chronic: Cat.1

Label elements and precautionary statement:

Pictogram:

Signal Word:
Danger

Hazard Statement:

H316	Causes mild skin irritation.
H330	Fatal if inhaled.
H301	Toxic if swallowed.
H371	May cause damage to organs (Central nervous system).
H373	May cause damage to organs (Central nervous system) through prolonged or repeated exposure.
H400	Very toxic to aquatic life.
H410	Very toxic to aquatic life with long lasting effects.

Precautionary Statement:

P101	If medical advice is needed, have product container or label at hand.
P102	Keep out of reach of children.
P103	Read carefully and follow all instructions.

Precautionary Statements (Prevention):

P271	Use only outdoors or in a well-ventilated area.
P260	Do not breathe dust/gas/mist/vapours.
P284	In case of inadequate ventilation wear respiratory protection.
P264	Wash contaminated body parts thoroughly after handling.
P270	Do not eat, drink or smoke when using this product.

Precautionary Statements (Response):

P310	Immediately call a POISON CENTER or physician.
P304 + P340	IF INHALED: Remove person to fresh air and keep comfortable for breathing.
P330	Rinse mouth.
P391	Collect spillage.

Precautionary Statements (Storage):

P403 + P233	Store in a well-ventilated place. Keep container tightly closed.
P405	Store locked up.

Precautionary Statements (Disposal):

P501	Dispose of contents and container to hazardous or special waste collection point.

Other hazards which do not result in classification:
See section 12 - Results of PBT and vPvB assessment.
If applicable information is provided in this section on other hazards which do not result in classification but which may contribute to the overall hazards of the substance or mixture.

May produce an allergic reaction. Contains:
1,2-benzisothiazol-3(2H)-one

3. Composition/information on ingredients

Chemical nature

Substance nature: mixture

crop protection product, insecticide, suspension concentrate (SC)

Hazardous ingredients

Chlorfenapyr

Content (W/W): 21.4 %	Acute Tox.: Cat. 2 (Inhalation - dust)
CAS Number: 122453-73-0	Acute Tox.: Cat. 3 (oral)
	STOT SE (Central nervous system): Cat. 2
	STOT RE (Central nervous system): Cat. 2
	Aquatic Acute: Cat. 1
	Aquatic Chronic: Cat. 1
	M-factor acute: 100
	M-factor chronic: 100

1,2-benzisothiazol-3(2H)-one

Content (W/W): < 0.05 %	Acute Tox.: Cat. 4 (oral)
CAS Number: 2634-33-5	Skin Corr./Irrit.: Cat. 2
	Eye Dam./Irrit.: Cat. 1
	Skin Sens.: Cat. 1
	Aquatic Acute: Cat. 1
	Aquatic Chronic: Cat. 1
	M-factor acute: 1
	M-factor chronic: 1

propane-1,2-diol
Content (W/W): < 10 %
CAS Number: 57-55-6

4. First-Aid Measures

General advice:
First aid personnel should pay attention to their own safety. If the patient is likely to become unconscious, place and transport in stable sideways position (recovery position). Immediately remove contaminated clothing.

If inhaled:
Keep patient calm, remove to fresh air, seek medical attention. In case of intoxication, call a poison control center or physician for treatment advice, taking the packaging or the label of the product.

On skin contact:
Immediately wash thoroughly with soap and water, seek medical attention. In case of intoxication, call a poison control center or physician for treatment advice, taking the packaging or the label of the product.

On contact with eyes:
Wash affected eyes for at least 15 minutes under running water with eyelids held open, consult an eye specialist.

On ingestion:
Immediately rinse mouth and then drink 200 - 300 ml water, do not induce vomiting, seek medical attention. If swallowed, seek medical advice immediately and show this container or label. Do not induce vomiting unless told to by a poison control center or doctor. Take patient to hospital immediately. In case of intoxication, call a poison control center or physician for treatment advice, taking the packaging or the label of the product. Medical monitoring for at least 7 days.

Note to physician:
Symptoms: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11., (Further) symptoms and / or effects are not known so far, Symptoms of poisoning may only appear after several hours or several days.
Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

5. Fire-Fighting Measures

Suitable extinguishing media:
water spray, carbon dioxide, foam, dry powder

Unsuitable extinguishing media for safety reasons:
water jet

Specific hazards:
carbon monoxide, carbon dioxide, hydrogen chloride, hydrogen bromide, nitrogen oxides, silicon oxides, halogenated compounds
The substances/groups of substances mentioned can be released in case of fire.

Special protective equipment:
Wear self-contained breathing apparatus and chemical-protective clothing.

Further information:
In case of fire and/or explosion do not breathe fumes. Keep containers cool by spraying with water if exposed to fire. Collect contaminated extinguishing water separately, do not allow to reach sewage or effluent systems. Dispose of fire debris and contaminated extinguishing water in accordance with official regulations.

6. Accidental Release Measures

Personal precautions:
Use personal protective clothing.Avoid contact with the skin, eyes and clothing.Do not breathe vapour/spray.

Environmental precautions:
Do not discharge into the subsoil/soil.Do not discharge into drains/surface waters/groundwater.

Methods for cleaning up or taking up:
For small amounts: Pick up with suitable absorbent material (e.g. sand, sawdust, general-purpose binder, kieselguhr).
For large amounts: Dike spillage. Pump off product.
Collect waste in suitable containers, which can be labeled and sealed. Clean contaminated floors and objects thoroughly with water and detergents, observing environmental regulations. Dispose of absorbed material in accordance with regulations. Wear suitable protective equipment.

7. Handling and Storage

Handling

Ensure thorough ventilation of stores and work areas. No special measures necessary if stored and handled correctly. When using do not eat, drink or smoke. Hands and/or face should be washed before breaks and at the end of the shift. Remove contaminated clothing and protective equipment before entering eating areas.

Protection against fire and explosion:
No special precautions necessary. The substance/product is non-combustible. Product is not explosive.

Storage

Segregate from foods and animal feeds.
Further information on storage conditions: Keep away from heat. Protect from direct sunlight.

Storage stability:
Storage duration: 36 Months

8. Exposure controls and personal protection

Components with occupational exposure limits

Chlorfenapyr, 122453-73-0;
TWA value 0.3 mg/m3 (BASF recomm. occupational exposure limit)

Personal protective equipment

Respiratory protection:
Suitable respiratory protection for lower concentrations or short-term effect: Combination filter for gases/vapours of organic, inorganic, acid inorganic, alkaline compounds and toxic particles (e. g. EN 14387 Type ABEK-P3)

Hand protection:
Suitable chemical resistant safety gloves (EN ISO 374-1) also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN ISO 374-1): E.g. nitrile rubber (0.4 mm), chloroprene rubber (0.5 mm), butyl rubber (0.7 mm) etc.

Eye protection:
Safety glasses with side-shields (frame goggles) (e.g. EN 166)

Body protection:
Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).

General safety and hygiene measures:
The statements on personal protective equipment in the instructions for use apply when handling crop-protection agents in final-consumer packing. Wearing of closed work clothing is recommended. Store work clothing separately. Keep away from food, drink and animal feeding stuffs.

9. Physical and Chemical Properties

Form:	liquid	
Colour:	white to light brown	
Odour:	sweetish, characteristic	
Odour threshold:	Not determined since toxic by inhalation.	
pH value:	approx. 6 - 8 (approx. 25 °C) (measured with the undiluted substance)	
Freezing point:	approx. 0 °C (1,013.3 hPa) Information applies to the solvent.	
Boiling point:	approx. 100 °C (1,013 mbar) Information applies to the solvent.	
Flash point:	Non-flammable.	(Regulation 440/2008/EC, A.9)
Evaporation rate:	not applicable	
Flammability (solid/gas):	not applicable	
Lower explosion limit:	As a result of our experience with this product and our knowledge of its composition we do not expect any hazard as long as the product is used appropriately and in accordance with the intended use.	
Upper explosion limit:	As a result of our experience with this product and our knowledge of its composition we do not expect any hazard as long as the product is used appropriately and in accordance with the intended use.	
Ignition temperature:	> 651 °C	(Regulation 440/2008/EC, A.15)

Thermal decomposition:	210 °C , 260 kJ/kg (onset temperature)	(DSC (OECD 113))
	Not a substance liable to self-decomposition according to UN transport regulations, class 4.1.	
Explosion hazard:	not explosive	
Fire promoting properties:	not fire-propagating	(Regulation 440/2008/EC, A.21)
Vapour pressure:	approx. 23.3 hPa (20 °C) Information applies to the solvent.	
Density:	approx. 1.13 g/cm3 (20 °C)	
Relative vapour density (air):	not applicable	
Solubility in water:	dispersible	
Partitioning coefficient n-octanol/water (log Pow):	not applicable	
Viscosity, dynamic:	approx. 757 mPa.s (20 °C)	

Other Information:
If necessary, information on other physical and chemical parameters is indicated in this section.

10. Stability and Reactivity

Conditions to avoid:
See SDS section 7 - Handling and storage.

Thermal decomposition:	210 °C, 260 kJ/kg (DSC (OECD 113)) (onset temperature)
Thermal decomposition:	Not a substance liable to self-decomposition according to UN transport regulations, class 4.1.

Substances to avoid:
strong acids, strong bases, strong oxidizing agents

Hazardous reactions:
No hazardous reactions if stored and handled as prescribed/indicated.

Hazardous decomposition products:
No hazardous decomposition products if stored and handled as prescribed/indicated.

Chemical stability:
The product is stable if stored and handled as prescribed/indicated.

Reactivity:

No hazardous reactions if stored and handled as prescribed/indicated.

11. Toxicological Information

Routes of exposure

Acute oral toxicity

Experimental/calculated data:
LD50mouse (oral): > 50 - < 300 mg/kg
The data on toxicology refer to the active ingredient. Same product LD50 is assigned for precautionary reasons in view of human poisoning incidents

Acute oral toxicity

LD50rat (oral): 560 - 567 mg/kg (OECD Guideline 423)

Acute inhalation toxicity

LC50 rat (by inhalation): 0.411 mg/l 4 h (OECD Guideline 403)
The product has not been tested. The statement has been derived from substances/products of a similar structure or composition. An aerosol was tested.

Acute dermal toxicity

LD50 rabbit (dermal): > 2,000 mg/kg
No mortality was observed.

Assessment of acute toxicity

Of high toxicity after short-term inhalation. Of pronounced toxicity after single ingestion. Virtually nontoxic after a single skin contact.

Information on: Chlorfenapyr

Acute oral toxicity

Experimental/calculated data:
LD50 mouse (oral): > 50 - < 300 mg/kg (OECD Guideline 401)

Symptoms

Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11.
(Further) symptoms and / or effects are not known so far Symptoms of poisoning may only appear after several hours or several days.

Irritation

Assessment of irritating effects:
Skin contact causes slight irritation. Not irritating to the eyes.

Experimental/calculated data:
Skin corrosion/irritation rabbit: Slightly irritating.
The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Serious eye damage/irritation rabbit: non-irritant

Respiratory/Skin sensitization

Assessment of sensitization:
There is no evidence of a skin-sensitizing potential.

Experimental/calculated data:
modified Buehler test guinea pig: Non-sensitizing. (OECD Guideline 406)

Germ cell mutagenicity

Assessment of mutagenicity:
Mutagenicity tests revealed no genotoxic potential. The product has not been tested. The statement has been derived from the properties of the individual components.

Carcinogenicity

Assessment of carcinogenicity:
The product has not been tested. The statement has been derived from the properties of the individual components. The results of various animal studies gave no indication of a carcinogenic effect.

Reproductive toxicity

Assessment of reproduction toxicity:
The product has not been tested. The statement has been derived from the properties of the individual components. The results of animal studies gave no indication of a fertility impairing effect.

Developmental toxicity

Assessment of teratogenicity:
The product has not been tested. The statement has been derived from the properties of the individual components. Animal studies gave no indication of a developmental toxic effect at doses that were not toxic to the parental animals.

Specific target organ toxicity (single exposure)

A single exposure may have relevant toxic effects on organs.

Target organ: Central nervous system
Remarks: The product has not been tested. The statement has been derived from the properties of the individual components.

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: Chlorfenapyr
Assessment of repeated dose toxicity:
Repeated exposure may affect certain organs. Affects the central nervous system.

Aspiration hazard

No aspiration hazard expected.

Other relevant toxicity information

Reported human health effects after incorporation: immediate symptoms after oral ingestion may comprise nausea, vomiting, sweating and abdominal pain. Subsequent symptoms emerging with a delay of a few days up to 10-14 days may include drowsiness, tachypnea, tachycardia or arrhythmia, hypertension, fever, diaphoresis, muscle fasciculation, muscle stiffness, weakness of limbs (sometimes leading to paralysis), hepatic and renal dysfunction, pancreatitis, and sudden disturbance of consciousness followed by coma and cardiac arrest. Misuse can be fatal.

12. Ecological Information

Ecotoxicity

Assessment of aquatic toxicity:
Very toxic to aquatic life with long lasting effects.
The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Toxicity to fish:
LC50 (96 h) 0.108 mg/l, Oncorhynchus mykiss (Flow through.)

Aquatic invertebrates:
EC50 (48 h) 0.0121 mg/l, Daphnia magna (static)

Aquatic plants:
EC50 (72 h) 4.09 mg/l, Selenastrum capricornutum (OECD Guideline 201)

Information on: Chlorfenapyr
Chronic toxicity to fish:
No observed effect concentration (93 d) 0.003678 mg/l, Oncorhynchus mykiss
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Information on: Chlorfenapyr
Chronic toxicity to aquatic invertebrates:
No observed effect concentration (28 d), 0.000172 mg/l, Mysidopsis bahia
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Mobility

Assessment transport between environmental compartments:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: Chlorfenapyr
Assessment transport between environmental compartments:
Following exposure to soil, adsorption to solid soil particles is probable, therefore contamination of groundwater is not expected.

Persistence and degradability

Assessment biodegradation and elimination (H2O):
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: Chlorfenapyr
Assessment biodegradation and elimination (H2O):
Not readily biodegradable (by OECD criteria).

Bioaccumulation potential

Assessment bioaccumulation potential:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: Chlorfenapyr
Bioaccumulation potential:
Bioconcentration factor: 116, Cyprinus carpio
Accumulation in organisms is not to be expected.

Additional information

Other ecotoxicological advice:
Do not discharge product into the environment without control.

13. Disposal Considerations

Must be sent to a suitable incineration plant, observing local regulations.

Contaminated packaging:
Contaminated packaging should be emptied as far as possible and disposed of in the same manner as the substance/product.

14. Transport Information

Domestic transport:

UN number or ID number:	UN 2902
UN proper shipping name:	PESTICIDE, LIQUID, TOXIC, N.O.S. (CHLORFENAPYR)
Transport hazard class(es):	6.1, EHSM
Packing group:	III
Environmental hazards:	yes
Special precautions for user:	None known

Sea transport
IMDG

UN number or ID number:	UN 2902
UN proper shipping name:	PESTICIDE, LIQUID, TOXIC, N.O.S. (CHLORFENAPYR)
Transport hazard class(es):	6.1, EHSM
Packing group:	III
Environmental hazards:	yes Marine pollutant: YES
Special precautions for user:	EmS: F-A; S-A

Air transport
IATA/ICAO

UN number or ID number:	UN 2902
UN proper shipping name:	PESTICIDE, LIQUID, TOXIC, N.O.S. (CHLORFENAPYR)
Transport hazard class(es):	6.1
Packing group:	III
Environmental hazards:	No Mark as dangerous for the environment is needed
Special precautions for user:	None known

15. Regulatory Information

Other regulations

To avoid risks to man and the environment, comply with the instructions for use.

16. Other Information

Vertical lines in the left hand margin indicate an amendment from the previous version.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.